REVTEX4

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Abstract

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I. ANTISYMMETRIC TENSOR PRODUCT

The antisymmetrizer is defined by

$$\mathcal{A}_N = \frac{1}{N!} \sum_{\sigma \in S_N} (-1)^{\pi(\sigma)} P_\sigma, \tag{1}$$

where

$$\begin{aligned} P_\sigma &: \bigotimes^N \mathcal{H}^1 \to \bigotimes^N \mathcal{H}^1 \\ P_\sigma \left|\varphi_{j_1} \otimes \cdots \otimes \varphi_{j_N}\right\rangle &= \left|\varphi_{j_{\sigma(1)}} \otimes \cdots \otimes \varphi_{j_{\sigma(N)}}\right\rangle . \end{aligned} \tag{2}$$

The antisymmetrizer is a self adjoint projection operator. The antisymmetric tensor product (wedge or Grassmann) is defined as

$$\left|\varphi_{j_1} \wedge \cdots \wedge \varphi_{j_N}\right\rangle = \mathcal{A}_N \left|\varphi_{j_1} \otimes \cdots \otimes \varphi_{j_N}\right\rangle = \frac{1}{N!} \left| \sum_{\sigma \in S_N} (-1)^{\pi(\sigma)} \varphi_{j_{\sigma(1)}} \otimes \cdots \otimes \varphi_{j_{\sigma(N)}} \right\rangle . \tag{3}$$

e.g.

$$\begin{aligned} \mathcal{A}_2 \left|\varphi_{j_1} \otimes \varphi_{j_2}\right\rangle &= \frac{1}{2} \left|\{\varphi_{j_1} \otimes \varphi_{j_2} - \varphi_{j_2} \otimes \varphi_{j_1}\}\right\rangle \\ \mathcal{A}_2^2 \left|\varphi_{j_1} \otimes \varphi_{j_2}\right\rangle &= \mathcal{A}_2 \frac{1}{2} \left|\{\varphi_{j_1} \otimes \varphi_{j_2} - \varphi_{j_2} \otimes \varphi_{j_1}\}\right\rangle \\ &= \left| \frac{1}{4} \{\varphi_{j_1} \otimes \varphi_{j_2} - \varphi_{j_2} \otimes \varphi_{j_1} - \varphi_{j_2} \otimes \varphi_{j_1} + \varphi_{j_1} \otimes \varphi_{j_2}\} \right\rangle \\ &= \frac{1}{2} \left|\{\varphi_{j_1} \otimes \varphi_{j_2} - \varphi_{j_2} \otimes \varphi_{j_1}\}\right\rangle \end{aligned} \tag{4}$$

and

$$
\mathcal{A}_3 \left|\varphi_{j_1} \otimes \varphi_{j_2} \otimes \varphi_{j_3}\right\rangle = \left|\varphi_{j_1} \wedge \varphi_{j_2} \wedge \varphi_{j_3}\right\rangle
$$
$$
= \frac{1}{6} \left|\{\varphi_{j_1} \otimes \varphi_{j_2} \otimes \varphi_{j_3} - \varphi_{j_2} \otimes \varphi_{j_1} \otimes \varphi_{j_3} - \varphi_{j_1} \otimes \varphi_{j_3} \otimes \varphi_{j_2} + \varphi_{j_3} \otimes \varphi_{j_2} \otimes \varphi_{j_2} + \varphi_{j_2} \otimes \varphi_{j_3} \otimes \varphi_{j_1} - \varphi_{j_2} \otimes \varphi_{j_3}\right. \tag{5}
$$

The vectors $\left|\varphi_{j_1} \wedge \cdots \wedge \varphi_{j_N}\right\rangle$ are however not normalized as e.g.

$$
\left\langle \varphi_{j_1} \wedge \varphi_{j_2}\right| \left.\varphi_{j_3} \wedge \varphi_{j_4}\right\rangle = \frac{1}{4} \left\langle \left(\varphi_{j_1} \otimes \varphi_{j_2} - \varphi_{j_2} \otimes \varphi_{j_1}\right)\right| \left.\left(\varphi_{j_3} \otimes \varphi_{j_4} - \varphi_{j_4} \otimes \varphi_{j_3}\right)\right\rangle
$$
$$
= \frac{1}{4} \left\{\left\langle \varphi_{j_1} \otimes \varphi_{j_2}\right| \left.\varphi_{j_3} \otimes \varphi_{j_4}\right\rangle - \left\langle \varphi_{j_1} \otimes \varphi_{j_2}\right| \left.\varphi_{j_4} \otimes \varphi_{j_3}\right\rangle - \left\langle \varphi_{j_2} \otimes \varphi_{j_1}\right| \left.\varphi_{j_3} \otimes \varphi_{j_4}\right\rangle + \left\langle \varphi_{j_2} \otimes \varphi_{j_1}\right| \left.\varphi_{j_4} \otimes \varphi_{j_3}\right\rangle\right\}
$$
$$
= \frac{1}{2} \left\{\left\langle \varphi_{j_1} \otimes \varphi_{j_2}\right| \left.\varphi_{j_3} \otimes \varphi_{j_4}\right\rangle - \left\langle \varphi_{j_1} \otimes \varphi_{j_2}\right| \left.\varphi_{j_4} \otimes \varphi_{j_3}\right\rangle\right\} = \frac{1}{2} \left\{\left\langle \varphi_{j_1}\right| \left.\varphi_{j_3}\right\rangle \left\langle \varphi_{j_2}\right| \left.\varphi_{j_4}\right\rangle - \left\langle \varphi_{j_1}\right| \left.\varphi_{j_4}\right\rangle \left\langle \varphi_{j_2}\right| \left.\varphi_{j_3}\right\rangle\right\}
$$
$$
= \frac{1}{2} \det \left\{ \begin{array}{cc} \left\langle \varphi_{j_1}\right| \left.\varphi_{j_3}\right\rangle & \left\langle \varphi_{j_1}\right| \left.\varphi_{j_4}\right\rangle \\ \left\langle \varphi_{j_2}\right| \left.\varphi_{j_3}\right\rangle & \left\langle \varphi_{j_2}\right| \left.\varphi_{j_4}\right\rangle \end{array} \right\} \tag{6}
$$

and one can define orthonormal ones as

$$
\left|\varphi_{j_1} \varphi_{j_2}\right\rangle \triangleq \sqrt{2} \left|\varphi_{j_1} \wedge \varphi_{j_2}\right\rangle, \tag{7}
$$

(note that the norm of $|\varphi_{j1}\varphi_{j2}\rangle$ is greater than the norm of $|\varphi_{j1} \wedge \varphi_{j2}\rangle$).$Thisgives$ $\left|\varphi_{j_1}\varphi_{j_2}\right\rangle =$ $\sqrt{2}\frac{1}{2}\left|\{\varphi_{j_1} \otimes \varphi_{j_2} - \varphi_{j_2} \otimes \varphi_{j_1}\}\right\rangle = \frac{1}{\sqrt{2}}\left|\{\varphi_{j_1} \otimes \varphi_{j_2} - \varphi_{j_2} \otimes \varphi_{j_1}\}\right\rangle$.(8)One should note

$$
\mathcal{A}_2 \frac{1}{\sqrt{2}} \left|\{\varphi_{j_1} \otimes \varphi_{j_2} - \varphi_{j_2} \otimes \varphi_{j_1}\}\right\rangle
$$
$$
= \frac{1}{\sqrt{2}} \left|\left\{\frac{1}{2}\left(\varphi_{j_1} \otimes \varphi_{j_2} - \varphi_{j_2} \otimes \varphi_{j_1}\right) - \frac{1}{2}\left(\varphi_{j_2} \otimes \varphi_{j_1} - \varphi_{j_1} \otimes \varphi_{j_2}\right)\right\}\right\rangle
$$
$$
= \frac{1}{\sqrt{2}} \left|\{\varphi_{j_1} \otimes \varphi_{j_2} - \varphi_{j_2} \otimes \varphi_{j_1}\}\right\rangle. \tag{9}
$$

As $\mathcal{A}_2$ projects onto $\mathcal{H}^2$ it must be true that

$$\begin{aligned}
\mathcal{A}_2 &= \sum_{1\le j_1<j_2<r} |\varphi_{j_1}\varphi_{j_2}\rangle\langle\varphi_{j_1}\varphi_{j_2}| = \frac{1}{8}\sum_{1\le j_1,j_2<r} |\{\varphi_{j_1}\otimes\varphi_{j_2}-\varphi_{j_2}\otimes\varphi_{j_1}\}\rangle\langle\{\varphi_{j_1}\otimes\varphi_{j_2}-\varphi_{j_2}\otimes\varphi_{j_1}\}| \\
&= \frac{1}{8}\sum_{1\le j_1,j_2<r} |\varphi_{j_1}\otimes\varphi_{j_2}\rangle\langle\varphi_{j_1}\otimes\varphi_{j_2}| - \frac{1}{8}\sum_{1\le j_1,j_2<r} |\varphi_{j_1}\otimes\varphi_{j_2}\rangle\langle\varphi_{j_2}\otimes\varphi_{j_1}| \\
&- \frac{1}{8}\sum_{1\le j_1,j_2<r} |\varphi_{j_2}\otimes\varphi_{j_1}\rangle\langle\varphi_{j_1}\otimes\varphi_{j_2}| + \frac{1}{8}\sum_{1\le j_1,j_2<r} |\varphi_{j_2}\otimes\varphi_{j_1}\rangle\langle\varphi_{j_2}\otimes\varphi_{j_1}| \\
&= \frac{1}{4}\sum_{1\le j_1,j_2<r} \{|\varphi_{j_1}\otimes\varphi_{j_2}\rangle\langle\varphi_{j_1}\otimes\varphi_{j_2}| - |\varphi_{j_2}\otimes\varphi_{j_1}\rangle\langle\varphi_{j_1}\otimes\varphi_{j_2}|\} \\
&= \frac{1}{2}\sum_{1\le j_1<j_2<r} \{|\varphi_{j_1}\otimes\varphi_{j_2}\rangle\langle\varphi_{j_1}\otimes\varphi_{j_2}| - |\varphi_{j_2}\otimes\varphi_{j_1}\rangle\langle\varphi_{j_1}\otimes\varphi_{j_2}|\} \\
&= \frac{1}{2}\sum_{1\le j_1<j_2<r} \{|(\varphi_{j_1}\otimes\varphi_{j_2}-\varphi_{j_2}\otimes\varphi_{j_1})\rangle\langle\varphi_{j_1}\otimes\varphi_{j_2}|\} = \sum_{1\le j_1<j_2<r} \{|\varphi_{j_1}\wedge\varphi_{j_2}\rangle\langle\varphi_{j_1}\otimes\varphi_{j_2}|\}\,.
\end{aligned} \tag{10}$$

In general

$$\langle\,\varphi_{j_1}\wedge\cdots\wedge\varphi_{j_N}|\,\varphi_{k_1}\wedge\cdots\wedge\varphi_{k_N}\rangle = \frac{1}{N!}\det\left\{\begin{array}{ccc} \langle\,\varphi_{j_1}|\,\varphi_{k_1}\rangle & \cdots & \langle\,\varphi_{j_1}|\,\varphi_{k_N}\rangle \\ \vdots & \ddots & \vdots \\ \langle\,\varphi_{j_N}|\,\varphi_{k_1}\rangle & \cdots & \langle\,\varphi_{j_N}|\,\varphi_{k_N}\rangle \end{array}\right\} \tag{11}$$

and one can define orthonormal vectors as

$$|\varphi_{j_1}\cdots\varphi_{j_N}\rangle \triangleq \sqrt{N!}\,|\varphi_{j_1}\wedge\cdots\wedge\varphi_{j_N}\rangle\,, \tag{12}$$

so that

$$|\varphi_{j_1}\cdots\varphi_{j_N}\rangle = \sqrt{N!}\frac{1}{N!}\left|\sum_{\sigma\in S_N}(-1)^{\pi(\sigma)}\,\varphi_{j_{\sigma(1)}}\otimes\cdots\otimes\varphi_{j_{\sigma(N)}}\right\rangle = \frac{1}{\sqrt{N!}}\left|\sum_{\sigma\in S_N}(-1)^{\pi(\sigma)}\,\varphi_{j_{\sigma(1)}}\otimes\cdots\otimes\varphi_{j_{\sigma(N)}}\right\rangle. \tag{13}$$

II. EXTERIOR PRODUCT OF THE IDENTITY

$$\begin{aligned}
P_{\mathcal{H}^1}\wedge P_{\mathcal{H}^1} &= \sum_{1\le j_1,k_1<r} |\varphi_{j_1}\rangle\langle\varphi_{j_1}|\wedge|\varphi_{k_1}\rangle\langle\varphi_{k_1}| = \sum_{1\le j_1,k_1<r} |\varphi_{j_1}\wedge\varphi_{k_1}\rangle\langle\varphi_{j_1}\wedge\varphi_{k_1}| \\
&= 2\sum_{1\le j_1<k_1<r} |\varphi_{j_1}\wedge\varphi_{k_1}\rangle\langle\varphi_{j_1}\wedge\varphi_{k_1}| = \sum_{1\le j_1<k_1<r} |\varphi_{j_1}\varphi_{k_1}\rangle\langle\varphi_{j_1}\varphi_{k_1}| = P_{\mathcal{H}^2}
\end{aligned} \tag{14}$$

III. EXPANSION MAP

The expansion map is defined as

$$\Gamma_P^N : \mathcal{B}\left(\mathcal{H}^P\right) \to \mathcal{B}\left(\mathcal{H}^N\right)$$
$$\Gamma_P^N\left(X\right) = \mathcal{A}_N X \otimes P_{\mathcal{H}^N} \mathcal{A}_N. \tag{15}$$

In particular

$$\Gamma_1^2 : \mathcal{B}\left(\mathcal{H}^1\right) \to \mathcal{B}\left(\mathcal{H}^2\right) \tag{16}$$

$\lceil$

$$\Gamma_1^2\left(X\right) = \mathcal{A}_2 X \otimes P_{\mathcal{H}^1} \mathcal{A}_2$$
$$= \frac{1}{4} \sum_{1\le j_1, j_2 < r} \left\{ \left|\varphi_{j_1} \otimes \varphi_{j_2}\right\rangle \left\langle \varphi_{j_1} \otimes \varphi_{j_2}\right| - \left|\varphi_{j_1} \otimes \varphi_{j_2}\right\rangle \left\langle \varphi_{j_2} \otimes \varphi_{j_1}\right| \right\} X \otimes P_{\mathcal{H}^1} \sum_{1\le k_1, k_2 < r} \left\{ \left|\varphi_{k_1} \otimes \varphi_{k_2}\right\rangle \left\langle \varphi_{k_1} \otimes \varphi_{k_2}\right| - \right.$$
$$= \frac{1}{4} \sum_{\substack{1\le j_1, j_2 < r \\ 1\le k_1, k_2 < r}} \left\{ \left|\varphi_{j_1} \otimes \varphi_{j_2}\right\rangle \left\langle \varphi_{k_1} \otimes \varphi_{k_2}\right| X_{j_1 k_1} \delta_{j_2 k_2} - \left|\varphi_{j_1} \otimes \varphi_{j_2}\right\rangle \left\langle \varphi_{k_2} \otimes \varphi_{k_1}\right| X_{j_1 k_2} \delta_{j_2 k_1} - \left|\varphi_{j_1} \otimes \varphi_{j_2}\right\rangle \left\langle \varphi_{k_1} \otimes \varphi_{k_2}\right| \right.$$
$$\left. + \left|\varphi_{j_1} \otimes \varphi_{j_2}\right\rangle \left\langle \varphi_{k_2} \otimes \varphi_{k_1}\right| X_{j_2 k_2} \delta_{j_1 k_1} \right\}$$
$$= \frac{1}{2} \sum_{\substack{1\le j_1, j_2 < r \\ 1\le k_1, k_2 < r}} \left|\varphi_{j_1} \otimes \varphi_{j_2}\right\rangle \left\langle \varphi_{k_1} \otimes \varphi_{k_2}\right| \left(X_{j_1 k_1} \delta_{j_2 k_2} - X_{j_2 k_1} \delta_{j_1 k_2}\right) - \left|\varphi_{j_1} \otimes \varphi_{j_2}\right\rangle \left\langle \varphi_{k_2} \otimes \varphi_{k_1}\right| \left(X_{j_1 k_2} \delta_{j_2 k_1} - X_{j_2 k_2} \delta_{j_1 \ldots}\right.$$
$$= \frac{1}{2} \sum_{\substack{1\le j_1, j_2 < r \\ 1\le k_1, k_2 < r}} \left|\varphi_{j_1} \otimes \varphi_{j_2}\right\rangle \left\langle \varphi_{k_1} \otimes \varphi_{k_2}\right| \left(X_{j_1 k_1} \delta_{j_2 k_2} - X_{j_2 k_1} \delta_{j_1 k_2}\right) - \left|\varphi_{j_2} \otimes \varphi_{j_1}\right\rangle \left\langle \varphi_{k_1} \otimes \varphi_{k_2}\right| \left(X_{j_2 k_1} \delta_{j_1 k_2} - X_{j_1 k_1} \delta_{j_2 \ldots}\right.$$
$$= \sum_{\substack{1\le j_1, j_2 < r \\ 1\le k_1, k_2 < r}} \left|\varphi_{j_1} \otimes \varphi_{j_2}\right\rangle \left\langle \varphi_{k_1} \otimes \varphi_{k_2}\right| \left(X_{j_1 k_1} \delta_{j_2 k_2} - X_{j_2 k_1} \delta_{j_1 k_2}\right) \tag{17}$$

$\rfloor$

$$\Gamma_1^2\left(X\right) = \mathcal{A}_2 X \otimes P_{\mathcal{H}^1} \mathcal{A}_2 = \mathcal{A}_2 \sum_{\substack{1\le j_1, j_2 < r \\ 1\le k_1 < r}} X_{j_1 j_2} \left|\varphi_{j_1}\right\rangle \left\langle \varphi_{j_2}\right| \otimes \left|\varphi_{k_1}\right\rangle \left\langle \varphi_{k_1}\right| \mathcal{A}_2$$
$$= \sum_{\substack{1\le j_1, j_2 < r \\ 1\le k_1 < r}} X_{j_1 j_2} \mathcal{A}_2 \left|\varphi_{j_1}\right\rangle \left\langle \varphi_{j_2}\right| \otimes \left|\varphi_{k_1}\right\rangle \left\langle \varphi_{k_1}\right| \mathcal{A}_2 = \sum_{\substack{1\le j_1, j_2 < r \\ 1\le k_1 < r}} X_{j_1 j_2} \mathcal{A}_2 \left|\varphi_{j_1} \otimes \varphi_{k_1}\right\rangle \left\langle \varphi_{j_2} \otimes \varphi_{k_1}\right| \mathcal{A}_2$$
$$= \sum_{\substack{1\le j_1, j_2 < r \\ 1\le k_1 < r}} X_{j_1 j_2} \left|\varphi_{j_1} \wedge \varphi_{k_1}\right\rangle \left\langle \varphi_{j_2} \wedge \varphi_{k_1}\right| = \sum_{\substack{1\le j_1, j_2 < r \\ 1\le k_1 < r}} X_{j_1 j_2} \left|\varphi_{j_1}\right\rangle \left\langle \varphi_{j_2}\right| \wedge \left|\varphi_{k_1}\right\rangle \left\langle \varphi_{k_1}\right|$$
$$= \sum_{\substack{1\le j_1, j_2 < r \\ 1\le k_1 < r}} X_{j_1 j_2} \left|\varphi_{j_1} \wedge \varphi_{k_1}\right\rangle \left\langle \varphi_{j_2} \wedge \varphi_{k_1}\right| = X \wedge P_{\mathcal{H}^1} = \frac{1}{2} \sum_{\substack{1\le j_1, j_2 < r \\ 1\le k_1 < r}} X_{j_1 j_2} \left|\varphi_{j_1} \varphi_{k_1}\right\rangle \left\langle \varphi_{j_2} \varphi_{k_1}\right| \tag{18}$$

i.e.

$$\Gamma_1^2(X) = \mathcal{A}_2 X \otimes P_{\mathcal{H}^1} \mathcal{A}_2 = X \wedge P_{\mathcal{H}^1} \tag{19}$$

and

$$\Gamma_1^2(P_{\mathcal{H}^1}) = \mathcal{A}_2 P_{\mathcal{H}^1} \otimes P_{\mathcal{H}^1} \mathcal{A}_2 = P_{\mathcal{H}^1} \wedge P_{\mathcal{H}^1}. \tag{20}$$

A. Restriction of SQ operators

Consider the restrictions of the number operator

$$\sum_{1\le l_1\le\infty} a_{l_1}^\dagger a_{l_1}\Big|_{\mathcal{H}^2} = \sum_{1\le l_1,j_1\le\infty} |\varphi_{l_1}\varphi_{j_1}\rangle\langle\varphi_{l_1}\varphi_{j_1}| = 2P_{\mathcal{H}^2} = 2P_{\mathcal{H}^1}\wedge P_{\mathcal{H}^1} = 2\Gamma_1^2(P_{\mathcal{H}^1}) \tag{21}$$

$$\sum_{1\le j_1\le\infty} a_{j_1}^\dagger a_{j_1}\Big|_{\mathcal{H}^3} = \sum_{1\le j_1,j_2,j_3\le r} |\varphi_{j_1}\varphi_{j_2}\varphi_{j_3}\rangle\langle\varphi_{j_1}\varphi_{j_2}\varphi_{j_3}| = 3P_{\mathcal{H}^3} = 3P_{\mathcal{H}^1}\wedge P_{\mathcal{H}^1}\wedge P_{\mathcal{H}^1} = 3\Gamma_1^3(P_{\mathcal{H}^1}). \tag{22}$$

⌈ In particular consider the restrictions of an occupation operator

$$\begin{aligned} a_1^\dagger a_1\Big|_{\mathcal{H}^2} &= \frac{1}{4}\sum_{1\le j_1,j_2\le\infty} |\varphi_{j_1}\varphi_{j_2}\rangle\langle\varphi_{j_1}\varphi_{j_2}|\, a_1^\dagger a_1 \sum_{1\le k_1,k_2\le\infty} |\varphi_{k_1}\varphi_{k_2}\rangle\langle\varphi_{k_1}\varphi_{k_2}| \\ &= \sum_{1\le j_1\le\infty} |\varphi_1\varphi_{j_1}\rangle\langle\varphi_1\varphi_{j_1}| = 2\sum_{1\le j_1\le\infty} |\varphi_1\wedge\varphi_{j_1}\rangle\langle\varphi_1\wedge\varphi_{j_1}| = 2|\varphi_1\rangle\langle\varphi_1|\wedge P_{\mathcal{H}^1} = 2\Gamma_1^2(|\varphi_1\rangle\langle\varphi_1|) \end{aligned} \tag{23}$$

and

$$\begin{aligned} a_1^\dagger a_1\Big|_{\mathcal{H}^3} &= \frac{1}{36}\sum_{1\le,j_1,j_2,j_3\le\infty} |\varphi_{j_1}\varphi_{j_2}\varphi_{j_3}\rangle\langle\varphi_{j_1}\varphi_{j_2}\varphi_{j_3}|\, a_1^\dagger a_1 \sum_{1\le j_1,j_2,j_3\le\infty} |\varphi_{j_1}\varphi_{j_2}\varphi_{j_3}\rangle\langle\varphi_{j_1}\varphi_{j_2}\varphi_{j_3}| \\ &= \frac{1}{4}\sum_{1\le j_2,j_3\le\infty} |\varphi_1\varphi_{j_2}\varphi_{j_3}\rangle\langle\varphi_1\varphi_{j_2}\varphi_{j_3}| = \frac{1}{2}\sum_{1\le j_2<j_3\le\infty} |\varphi_1\varphi_{j_2}\varphi_{j_3}\rangle\langle\varphi_1\varphi_{j_2}\varphi_{j_3}| \\ &= \frac{3}{4}\sum_{1\le j_2,j_3\le\infty} |\varphi_1\wedge\varphi_{j_2}\wedge\varphi_{j_3}\rangle\langle\varphi_1\wedge\varphi_{j_2}\wedge\varphi_{j_3}| \qquad (24) \\ &= \frac{3}{4}\sum_{1\le j_2,j_3\le\infty} |\varphi_1\rangle\langle\varphi_1|\wedge|\varphi_{j_2}\wedge\varphi_{j_3}\rangle\langle\varphi_{j_2}\wedge\varphi_{j_3}| = \frac{3}{4}\sum_{1\le j_2,j_3\le\infty} |\varphi_1\rangle\langle\varphi_1|\wedge P_{\mathcal{H}^1}\wedge P_{\mathcal{H}^1} \\ &= \frac{3}{2}\sum_{1\le j_2,j_3\le\infty} |\varphi_1\rangle\langle\varphi_1|\wedge|\varphi_{j_2}\varphi_{j_3}\rangle\langle\varphi_{j_2}\varphi_{j_3}| = 3\sum_{1\le j_2<j_3\le\infty} |\varphi_1\rangle\langle\varphi_1|\wedge|\varphi_{j_2}\varphi_{j_3}\rangle\langle\varphi_{j_2}\varphi_{j_3}| \\ &= 3|\varphi_1\rangle\langle\varphi_1|\wedge P_{\mathcal{H}^2} = 3\Gamma_1^3(|\varphi_1\rangle\langle\varphi_1|). \qquad (25) \end{aligned}$$

as

$$\left|\varphi_{j_1}\varphi_{j_2}\right\rangle \triangleq \sqrt{2}\left|\varphi_{j_1}\wedge\varphi_{j_2}\right\rangle, \tag{26}$$

Thus one can observe *the important relationships*

$$\begin{aligned}\Gamma_1^2\left(\left|\varphi_1\right\rangle\left\langle\varphi_1\right|\right) = \left|\varphi_1\right\rangle\left\langle\varphi_1\right| \wedge P_{\mathcal{H}^1} = \frac{1}{2}\sum_{1\le j_1\le\infty}\left|\varphi_1\varphi_{j_1}\right\rangle\left\langle\varphi_1\varphi_{j_1}\right| \\ \sum_{1\le j_1\le\infty}\left|\varphi_1\varphi_{j_1}\right\rangle\left\langle\varphi_1\varphi_{j_1}\right| = 2\Gamma_1^2\left(\left|\varphi_1\right\rangle\left\langle\varphi_1\right|\right)\end{aligned} \tag{27}$$

giving

$$\begin{aligned}\Gamma_1^2\left(\left|\varphi_1\right\rangle\left\langle\varphi_1\right|\right) = \frac{1}{2}\sum_{1\le j_1\le\infty}\left|\varphi_1\varphi_{j_1}\right\rangle\left\langle\varphi_1\varphi_{j_1}\right| \\ \sum_{1\le j_1\le\infty}\left|\varphi_1\varphi_{j_1}\right\rangle\left\langle\varphi_1\varphi_{j_1}\right| = 2\Gamma_1^2\left(\left|\varphi_1\right\rangle\left\langle\varphi_1\right|\right)\end{aligned} \tag{28}$$

and

$$\Gamma_1^3\left(\left|\varphi_1\right\rangle\left\langle\varphi_1\right|\right) = \left|\varphi_1\right\rangle\left\langle\varphi_1\right| \wedge P_{\mathcal{H}^2} = \sum_{1\le j_2<j_3\le\infty}\left|\varphi_1\right\rangle\left\langle\varphi_1\right| \wedge \left|\varphi_{j_2}\varphi_{j_3}\right\rangle\left\langle\varphi_{j_2}\varphi_{j_3}\right| = 2\sum_{1\le j_2,j_3\le\infty}\left|\varphi_1\right\rangle\left\langle\varphi_1\right| \wedge \left|\varphi_{j_2}\right\rangle\left\langle\varphi_{j_2}\right| \wedge \left|\varphi_j\right.$$

giving

$$\Gamma_1^3\left(\left|\varphi_1\right\rangle\left\langle\varphi_1\right|\right) = \frac{1}{3}\sum_{1\le j_2<j_3\le\infty}\left|\varphi_1\varphi_{j_2}\varphi_{j_3}\right\rangle\left\langle\varphi_1\varphi_{j_2}\varphi_{j_3}\right| \tag{29}$$

$$\sum_{1\le j_2<j_3\le\infty}\left|\varphi_1\varphi_{j_2}\varphi_{j_3}\right\rangle\left\langle\varphi_1\varphi_{j_2}\varphi_{j_3}\right| = 3\Gamma_1^3\left(\left|\varphi_1\right\rangle\left\langle\varphi_1\right|\right) \tag{30}$$

leading to

$$\begin{aligned}\Gamma_1^N\left(\left|\varphi_1\right\rangle\left\langle\varphi_1\right|\right) = \frac{1}{N}\left|\varphi_1\right\rangle\left\langle\varphi_1\right| \wedge P_{\mathcal{H}^{N-1}} \\ \left|\varphi_1\right\rangle\left\langle\varphi_1\right| \wedge P_{\mathcal{H}^{N-1}} = N\Gamma_1^N\left(\left|\varphi_1\right\rangle\left\langle\varphi_1\right|\right)\end{aligned} \tag{31}$$

i.e.

$$P_{\mathcal{H}^N} = N\Gamma_1^N\left(P_{\mathcal{H}^1}\right). \tag{32}$$

⌋

Thus

$$\begin{aligned}\sum_{1\le l_1\le\infty} a_{l_1}^\dagger a_{l_1} &= P_{\mathcal{H}^1} \wedge \left|\phi\right\rangle\left\langle\phi\right| + 2P_{\mathcal{H}^1} \wedge P_{\mathcal{H}^1} + 3P_{\mathcal{H}^1} \wedge P_{\mathcal{H}^2} + \cdots \\ &= \Gamma_1^1\left(P_{\mathcal{H}^1}\right) + \Gamma_1^2\left(P_{\mathcal{H}^1}\right) + \Gamma_1^3\left(P_{\mathcal{H}^1}\right) + \Gamma_1^4\left(P_{\mathcal{H}^1}\right) + \cdots \\ &= P_{\mathcal{H}^1} \wedge \mathfrak{S}_1\end{aligned} \tag{33}$$

$$\begin{aligned} a_1^\dagger a_1 &= |\varphi_1\rangle\langle\varphi_1| + |\varphi_1\varphi_2\rangle\langle\varphi_1\varphi_2| + |\varphi_1\varphi_3\rangle\langle\varphi_1\varphi_3| + |\varphi_1\varphi_2\varphi_3\rangle\langle\varphi_1\varphi_2\varphi_3| \\ &= |\varphi_1\rangle\langle\varphi_1| + \Gamma_1^2\left(|\varphi_1\rangle\langle\varphi_1|\right) + \Gamma_1^3\left(|\varphi_1\rangle\langle\varphi_1|\right). \end{aligned} \tag{34}$$

$$\begin{aligned} a_2^\dagger a_2 &= |\varphi_2\rangle\langle\varphi_2| + |\varphi_1\varphi_2\rangle\langle\varphi_1\varphi_2| + |\varphi_2\varphi_3\rangle\langle\varphi_2\varphi_3| + |\varphi_1\varphi_2\varphi_3\rangle\langle\varphi_1\varphi_2\varphi_3| \\ &= |\varphi_2\rangle\langle\varphi_2| + \Gamma_1^2\left(|\varphi_2\rangle\langle\varphi_2|\right) + \Gamma_1^3\left(|\varphi_2\rangle\langle\varphi_2|\right). \end{aligned} \tag{35}$$

$$\begin{aligned} a_3^\dagger a_3 &= |\varphi_3\rangle\langle\varphi_3| + |\varphi_1\varphi_3\rangle\langle\varphi_1\varphi_3| + |\varphi_2\varphi_3\rangle\langle\varphi_2\varphi_3| + |\varphi_1\varphi_2\varphi_3\rangle\langle\varphi_1\varphi_2\varphi_3| \\ &= |\varphi_3\rangle\langle\varphi_3| + \Gamma_1^2\left(|\varphi_3\rangle\langle\varphi_3|\right) + \Gamma_1^3\left(|\varphi_3\rangle\langle\varphi_3|\right). \end{aligned} \tag{36}$$

$$a_1^\dagger a_1 + a_2^\dagger a_2 + a_3^\dagger a_3 = P_{\mathcal{H}^1} + 2P_{\mathcal{H}^2} + 3P_{\mathcal{H}^3} \tag{37}$$

$$a_1^\dagger a_1 + a_2^\dagger a_2 + a_3^\dagger a_3 = P_{\mathcal{H}^1} + 2P_{\mathcal{H}^2} + 3P_{\mathcal{H}^3} \tag{38}$$

where

$$\mathfrak{S}_1 = \left\{ \sum_{0 \le N \le r} \binom{N+1}{1} P_{\mathcal{H}^N} \right\} = |\phi\rangle\langle\phi| + 2P_{\mathcal{H}^1} + 3P_{\mathcal{H}^1} \wedge P_{\mathcal{H}^1} + 4P_{\mathcal{H}^1} \wedge P_{\mathcal{H}^1} \wedge P_{\mathcal{H}^1} + \cdots. \tag{39}$$

One can check that

$$\left(P_{\mathcal{H}^1}\right)^{3\wedge} = P_{\mathcal{H}^1} \wedge P_{\mathcal{H}^1} \wedge P_{\mathcal{H}^1} = P_{\mathcal{H}^2} \wedge P_{\mathcal{H}^1} = P_{\mathcal{H}^1} \wedge P_{\mathcal{H}^2} = P_{\mathcal{H}^3} \tag{40}$$

by noting

$$\begin{aligned} P_{\mathcal{H}^3} &= \sum_{1 \le j_1 < j_2 < j_3 \le r} |\varphi_{j_1}\varphi_{j_2}\varphi_{j_3}\rangle\langle\varphi_{j_1}\varphi_{j_2}\varphi_{j_3}| = \frac{1}{6} \sum_{1 \le j_1, j_2, j_3 \le r} |\varphi_{j_1}\varphi_{j_2}\varphi_{j_3}\rangle\langle\varphi_{j_1}\varphi_{j_2}\varphi_{j_3}| \\ &= \sum_{1 \le j_1, j_2, j_3 \le r} |\varphi_{j_1} \wedge \varphi_{j_2} \wedge \varphi_{j_3}\rangle\langle\varphi_{j_1} \wedge \varphi_{j_2} \wedge \varphi_{j_3}| = \sum_{1 \le j_1, j_2, j_3 \le r} |\varphi_{j_1}\rangle\langle\varphi_{j_1}| \wedge |\varphi_{j_2}\rangle\langle\varphi_{j_2}| \wedge |\varphi_{j_3}\rangle\langle\varphi_{j_3}| = P_{\mathcal{H}^1} \wedge \end{aligned} \tag{41}$$

Thus

$$I_{\mathcal{H}_\mathcal{F}} = \sum_{0 \le N \le r} P_{\mathcal{H}^N} = \sum_{0 \le N \le r} \left(P_{\mathcal{H}^1}\right)^{N\wedge} = |\phi\rangle\langle\phi| + P_{\mathcal{H}^1} + P_{\mathcal{H}^1} \wedge P_{\mathcal{H}^1} + \cdots. \tag{42}$$

The general expression for the expansion map $\mathcal{B}\left(\mathcal{H}^P\right) \to \mathcal{B}\left(\mathcal{H}^N\right)$ is

$$\begin{aligned} &\Gamma_P^N : \mathcal{B}\left(\mathcal{H}^P\right) \to \mathcal{B}\left(\mathcal{H}^N\right) \\ &\Gamma_P^N(X) = \mathcal{A}_N X \otimes P_{\mathcal{H}^{N-P}} \mathcal{A}_N = X \wedge P_{\mathcal{H}^{N-P}} \end{aligned} \tag{43}$$

and in complete generality

$$\mathcal{B}\left(\mathcal{H}^P,\mathcal{H}^Q\right) \to \mathcal{B}\left(\mathcal{H}^N,\mathcal{H}^M\right)$$
$$\Gamma^{NM}_{PQ}\left(X\right) = \mathcal{A}_N X \otimes P_{\mathcal{H}^{N-P},\mathcal{H}^{M-Q}}\mathcal{A}_M = X \wedge P_{\mathcal{H}^{N-P},\mathcal{H}^{M-Q}}, \tag{44}$$

where

$$P_{\mathcal{H}^{N-P},\mathcal{H}^{M-Q}} = \sum_{\substack{1\le j_1\cdots<j_{N-P}\le r\\ 1\le k_1\cdots<k_{M-Q}\le r}} \left|\varphi_{j_1}\cdots\varphi_{j_{N-P}}\right\rangle\left\langle\varphi_{k_1}\cdots\varphi_{k_{M-Q}}\right| \tag{45}$$

IV. SECOND QUANTIZATION MAP

By considering the restrictions of $a_j^\dagger$ to $\mathcal{H}^N$, i.e. $a_j^\dagger P_{\mathcal{H}^N}$, for $0\le N\le\infty$ one obtains

$$\begin{aligned}
a_j^\dagger &= \left|\varphi_j\right\rangle\left\langle\phi\right| + \sum_{1\le N\le\infty}\sum_{1\le j_1<\cdots<j_N\le\infty}\left|\varphi_j\varphi_{j_1}\cdots\varphi_{j_N}\right\rangle\left\langle\varphi_{j_1}\cdots\varphi_{j_N}\right| \\
&= \sum_{0\le N\le\infty}\sum_{1\le j_1<\cdots<j_N\le\infty}\left|\varphi_j\varphi_{j_1}\cdots\varphi_{j_N}\right\rangle\left\langle\varphi_{j_1}\cdots\varphi_{j_N}\right| \\
&= \sum_{0\le N\le\infty}\sum_{1\le j_1,\cdots,j_N\le\infty}\left(N!\right)^{-1}\left|\varphi_j\varphi_{j_1}\cdots\varphi_{j_N}\right\rangle\left\langle\varphi_{j_1}\cdots\varphi_{j_N}\right| \\
&= \sum_{0\le N\le\infty}\sum_{1\le j_1,\cdots,j_N\le\infty}\sqrt{(N+1)}\left|\varphi_j\wedge\varphi_{j_1}\wedge\cdots\wedge\varphi_{j_N}\right\rangle\left\langle\varphi_{j_1}\wedge\cdots\wedge\varphi_{j_N}\right|
\end{aligned} \tag{46}$$

as

$$\begin{aligned}
\left(N!\right)^{-1}\left|\varphi_j\varphi_{j_1}\cdots\varphi_{j_N}\right\rangle\left\langle\varphi_{j_1}\cdots\varphi_{j_N}\right| &= \left(N!\right)^{-1}\sqrt{(N+1)!}\left|\varphi_j\wedge\varphi_{j_1}\wedge\cdots\wedge\varphi_{j_N}\right\rangle\left\langle\varphi_{j_1}\cdots\varphi_{j_N}\right| \\
&= \left(\sqrt{N!}\right)^{-1}\sqrt{(N+1)}\left|\varphi_j\wedge\varphi_{j_1}\wedge\cdots\wedge\varphi_{j_N}\right\rangle\left\langle\varphi_{j_1}\cdots\varphi_{j_N}\right|.
\end{aligned} \tag{47}$$

Thus one can define the second quantization map Γ

$$\Gamma:\mathcal{F}\to\mathcal{F} \tag{48}$$

from the Fock algebra to itself by

$$\begin{aligned}
\Gamma\left(\left|\varphi_j\right\rangle\left\langle\phi\right|\right) = a_j^\dagger &= \sum_{0\le N\le\infty}\sum_{1\le j_1<\cdots<j_N\le\infty}\left|\varphi_j\varphi_{j_1}\cdots\varphi_{j_N}\right\rangle\left\langle\varphi_{j_1}\cdots\varphi_{j_N}\right| \\
&= \sum_{0\le N\le\infty}\sum_{1\le j_1,\cdots,j_N\le\infty}\sqrt{(N+1)}\left|\varphi_j\wedge\varphi_{j_1}\wedge\cdots\wedge\varphi_{j_N}\right\rangle\left\langle\varphi_{j_1}\wedge\cdots\wedge\varphi_{j_N}\right|.
\end{aligned} \tag{49}$$

The action on the vacuum state is thus given by

$$\Gamma\left(|\phi\rangle\langle\phi|\right)=\sum_{0\le N\le\infty}\sum_{1\le j_1<\cdots<j_N\le\infty}|\varphi_{j_1}\cdots\varphi_{j_N}\rangle\langle\varphi_{j_1}\cdots\varphi_{j_N}|=I_{\mathcal{H}_{\mathcal{F}}}\tag{50}$$

i.e. $I_{\mathcal{H}_{\mathcal{F}}}$ is a bases vector in the transformed basis!

Hence

$$\begin{aligned}a_{k_1}^{\dagger}a_{k_2}^{\dagger}&=|\varphi_{k_1}\varphi_{k_2}\rangle\langle\phi|+\sum_{1\le N\le\infty}\sum_{1\le j_1<\cdots<j_N\le\infty}|\varphi_{k_1}\varphi_{k_2}\varphi_{j_1}\cdots\varphi_{j_N}\rangle\langle\varphi_{j_1}\cdots\varphi_{j_N}|\\&=\sum_{0\le N\le\infty}\sum_{1\le j_1<\cdots<j_N\le\infty}|\varphi_{k_1}\varphi_{k_2}\varphi_{j_1}\cdots\varphi_{j_N}\rangle\langle\varphi_{j_1}\cdots\varphi_{j_N}|\\&=\sum_{0\le N\le\infty}\sum_{1\le j_1,\cdots,j_N\le\infty}\sqrt{(N+1)(N+2)}\,|\varphi_{k_1}\wedge\varphi_{k_2}\wedge\varphi_{j_1}\wedge\cdots\wedge\varphi_{j_N}\rangle\langle\varphi_{j_1}\wedge\cdots\wedge\varphi_{j_N}|\,.\end{aligned}\tag{51}$$

and in general

$$\begin{aligned}a_{k_1}^{\dagger}\cdots a_{k_P}^{\dagger}&=|\varphi_{k_1}\cdots\varphi_{k_P}\rangle\langle\phi|+\sum_{1\le N\le\infty}\sum_{1\le j_1<\cdots<j_N\le\infty}|\varphi_{k_1}\cdots\varphi_{k_P}\varphi_{j_1}\cdots\varphi_{j_N}\rangle\langle\varphi_{j_1}\cdots\varphi_{j_N}|\\&=\sum_{0\le N\le\infty}\sum_{1\le j_1<\cdots<j_N\le\infty}|\varphi_{k_1}\cdots\varphi_{k_P}\varphi_{j_1}\cdots\varphi_{j_N}\rangle\langle\varphi_{j_1}\cdots\varphi_{j_N}|\\&=\sum_{0\le N\le\infty}\sum_{1\le j_1,\cdots,j_N\le\infty}\frac{\sqrt{(N+P)\cdots(N+1)}}{\sqrt{P!}}\,|\varphi_{k_1}\wedge\cdots\wedge\varphi_{k_p}\wedge\varphi_{j_1}\wedge\cdots\wedge\varphi_{j_N}\rangle\langle\varphi_{j_1}\wedge\cdots\wedge\varphi_{j_N}|\\&=\sum_{0\le N\le\infty}\sum_{1\le j_1,\cdots,j_N\le\infty}\sqrt{\binom{N+P}{P}}\,|\varphi_{k_1}\wedge\cdots\wedge\varphi_{k_P}\wedge\varphi_{j_1}\wedge\cdots\wedge\varphi_{j_N}\rangle\langle\varphi_{j_1}\wedge\cdots\wedge\varphi_{j_N}|\,.\end{aligned}\tag{52}$$

Annihilators can be expressed in an analogous fashion by

$$\begin{aligned}a_j&=|\phi\rangle\langle\varphi_j|+\sum_{1\le N\le\infty}\sum_{1\le j_1<\cdots<j_N\le\infty}|\varphi_{j_1}\cdots\varphi_{j_N}\rangle\langle\varphi_j\varphi_{j_1}\cdots\varphi_{j_N}|\\&=\sum_{0\le N\le\infty}\sum_{1\le j_1<\cdots<j_N\le\infty}|\varphi_{j_1}\cdots\varphi_{j_N}\rangle\langle\varphi_j\varphi_{j_1}\cdots\varphi_{j_N}|\\&=\sum_{0\le N\le\infty}\sum_{1\le j_1,\cdots,j_N\le\infty}\sqrt{(N+1)}\,|\varphi_{j_1}\wedge\cdots\wedge\varphi_{j_N}\rangle\langle\varphi_j\wedge\varphi_{j_1}\wedge\cdots\wedge\varphi_{j_N}|\,.\end{aligned}\tag{53}$$

Hence

$$a_{k_2}a_{k_1}=|\phi\rangle\langle\varphi_{k_1}\varphi_{k_2}|+\sum_{1\le N\le\infty}\sum_{1\le j_1<\cdots<j_N\le\infty}|\varphi_{j_1}\cdots\varphi_{j_N}\rangle\langle\varphi_{k_1}\varphi_{k_2}\varphi_{j_1}\cdots\varphi_{j_N}|$$
$$=\sum_{0\le N\le\infty}\sum_{1\le j_1<\cdots<j_N\le\infty}|\varphi_{j_1}\cdots\varphi_{j_N}\rangle\langle\varphi_{k_1}\varphi_{k_2}\varphi_{j_1}\cdots\varphi_{j_N}|$$
$$=\sum_{0\le N\le\infty}\sum_{1\le j_1,\cdots,j_N\le\infty}\sqrt{(N+1)(N+2)}\,|\varphi_{j_1}\wedge\cdots\wedge\varphi_{j_N}\rangle\langle\varphi_{k_1}\wedge\varphi_{k_2}\wedge\varphi_{j_1}\wedge\cdots\wedge\varphi_{j_N}|\,. \tag{54}$$

and

$$a_{k_Q}\cdots a_{k_1}=|\phi\rangle\langle\varphi_{k_1}\cdots\varphi_{k_Q}|+\sum_{1\le N\le\infty}\sum_{1\le j_1<\cdots<j_N\le\infty}|\varphi_{j_1}\cdots\varphi_{j_N}\rangle\langle\varphi_{k_1}\cdots\varphi_{k_Q}\varphi_{j_1}\cdots\varphi_{j_N}|$$
$$=\sum_{0\le N\le\infty}\sum_{1\le j_1<\cdots<j_N\le\infty}|\varphi_{j_1}\cdots\varphi_{j_N}\rangle\langle\varphi_{k_1}\cdots\varphi_{k_Q}\varphi_{j_1}\cdots\varphi_{j_N}|$$
$$=\sum_{0\le N\le\infty}\sum_{1\le j_1,\cdots,j_N\le\infty}\frac{\sqrt{(N+Q)(N+1)}}{\sqrt{Q!}}|\varphi_{j_1}\wedge\cdots\wedge\varphi_{j_N}\rangle\langle\varphi_{k_1}\wedge\cdots\wedge\varphi_{k_Q}\wedge\varphi_{j_1}\wedge\cdots\wedge\varphi_{j_N}|$$
$$=\sum_{0\le N\le\infty}\sum_{1\le j_1,\cdots,j_N\le\infty}\sqrt{\binom{N+Q}{Q}}|\varphi_{j_1}\wedge\cdots\wedge\varphi_{j_N}\rangle\langle\varphi_{k_1}\wedge\cdots\wedge\varphi_{k_Q}\wedge\varphi_{j_1}\wedge\cdots\wedge\varphi_{j_N}|\,. \tag{55}$$

One particle operators can be expressed as

$$a^\dagger_{l_1}a_{m_1}=\Gamma(|\varphi_{l_1}\rangle\langle\varphi_{m_1}|)$$
$$=\sum_{\substack{0\le N,M\le\infty\\1\le j_1<\cdots<j_N\le\infty\\1\le k_1<\cdots<k_M\le\infty}}|\varphi_{l_1}\varphi_{j_1}\cdots\varphi_{j_N}\rangle\langle\varphi_{j_1}\cdots\varphi_{j_N}|\varphi_{k_1}\cdots\varphi_{k_M}\rangle\langle\varphi_{m_1}\varphi_{k_1}\cdots\varphi_{k_M}|$$
$$=\sum_{0\le N\le\infty}\sum_{1\le j_1<\cdots<j_N\le\infty}|\varphi_{l_1}\varphi_{j_1}\cdots\varphi_{j_N}\rangle\langle\varphi_{m_1}\varphi_{j_1}\cdots\varphi_{j_N}|$$
$$=\sum_{0\le N\le\infty}\sum_{1\le j_1,\cdots,j_N\le\infty}(N+1)|\varphi_{l_1}\wedge\varphi_{j_1}\wedge\cdots\wedge\varphi_{j_N}\rangle\langle\varphi_{m_1}\wedge\varphi_{j_1}\wedge\cdots\wedge\varphi_{j_N}|=\sum_{0\le N\le\infty}(N+1)|\varphi_{l_1}\rangle\langle\varphi_{m_1}|\wedge$$
$$=|\varphi_{l_1}\rangle\langle\varphi_{m_1}|+2|\varphi_{l_1}\rangle\langle\varphi_{m_1}|\wedge P_{\mathcal{H}^1}+3|\varphi_{l_1}\rangle\langle\varphi_{m_1}|\wedge P_{\mathcal{H}^2}+\cdots=\Gamma^1_1(|\varphi_{l_1}\rangle\langle\varphi_{m_1}|)+2\Gamma^2_1(|\varphi_{l_1}\rangle\langle\varphi_{m_1}|)+\cdots$$
$$=\sum_{1\le N\le\infty}N\Gamma^N_1(|\varphi_{l_1}\rangle\langle\varphi_{m_1}|)=\sum_{1\le N\le\infty}\binom{N}{1}\Gamma^N_1(|\varphi_{l_1}\rangle\langle\varphi_{m_1}|)=\Gamma_1(|\varphi_{l_1}\rangle\langle\varphi_{m_1}|) \tag{56}$$

and

$$
\begin{aligned}
&a_{l_1}^{\dagger} a_{l_2}^{\dagger} a_{m_2} a_{m_1} = \Gamma \left(\left| \varphi_{l_1} \varphi_{l_2} \right\rangle \left\langle \varphi_{m_1} \varphi_{m_2} \right| \right) \\
&= \sum_{\substack{0 \leq N, M \leq \infty \\ 1 \leq j_1 < \cdots < j_N \leq \infty \\ 1 \leq k_1 < \cdots < k_M \leq \infty}} \left| \varphi_{l_1} \varphi_{l_2} \varphi_{j_1} \cdots \varphi_{j_N} \right\rangle \left\langle \varphi_{j_1} \cdots \varphi_{j_N} \right| \varphi_{k_1} \cdots \varphi_{k_M} \left\rangle \right\langle \varphi_{m_1} \varphi_{m_2} \varphi_{k_1} \cdots \varphi_{k_M} \right| \\
&= \sum_{0 \leq N \leq \infty} \sum_{1 \leq j_1 < \cdots < j_N \leq \infty} \left| \varphi_{l_1} \varphi_{l_2} \varphi_{j_1} \cdots \varphi_{j_N} \right\rangle \left\langle \varphi_{m_1} \varphi_{m_2} \varphi_{j_1} \cdots \varphi_{j_N} \right| \\
&= \sum_{0 \leq N \leq \infty} \sum_{1 \leq j_1, \cdots, j_N \leq \infty} (N+1)(N+2) \left| \varphi_{l_1} \wedge \varphi_{l_2} \wedge \varphi_{j_1} \wedge \cdots \wedge \varphi_{j_N} \right\rangle \left\langle \varphi_{m_1} \wedge \varphi_{m_2} \wedge \varphi_{j_1} \wedge \cdots \wedge \varphi_{j_N} \right| \\
&= \sum_{0 \leq N \leq \infty} (N+1)(N+2) \left| \varphi_{l_1} \right\rangle \left\langle \varphi_{m_1} \right| \wedge \left| \varphi_{l_2} \right\rangle \left\langle \varphi_{m_2} \right| \wedge P_{\mathcal{H}^N} = \sum_{0 \leq N \leq \infty} \frac{(N+1)(N+2)}{2} \left| \varphi_{l_1} \varphi_{l_2} \right\rangle \left\langle \varphi_{m_1} \varphi_{m_2} \right| \wedge P_{\mathcal{H}} \\
&= \sum_{0 \leq N \leq \infty} \frac{(N+1)(N+2)}{2} \Gamma_2^N \left(\left| \varphi_{l_1} \varphi_{l_2} \right\rangle \left\langle \varphi_{m_1} \varphi_{m_2} \right| \right) = \left| \varphi_{l_1} \varphi_{l_2} \right\rangle \left\langle \varphi_{m_1} \varphi_{m_2} \right| + 3 \Gamma_2^3 \left(\left| \varphi_{l_1} \varphi_{l_2} \right\rangle \left\langle \varphi_{m_1} \varphi_{m_2} \right| \right) + 6 \Gamma_2^4 (\left| \varphi_{l_1} \right. \\
&= \binom{2}{2} \left| \varphi_{l_1} \varphi_{l_2} \right\rangle \left\langle \varphi_{m_1} \varphi_{m_2} \right| + \binom{3}{2} \Gamma_2^3 \left(\left| \varphi_{l_1} \varphi_{l_2} \right\rangle \left\langle \varphi_{m_1} \varphi_{m_2} \right| \right) + \binom{4}{2} \Gamma_2^4 \left(\left| \varphi_{l_1} \varphi_{l_2} \right\rangle \left\langle \varphi_{m_1} \varphi_{m_2} \right| \right) + \cdots \\
&= \sum_{2 \leq N \leq \infty} \binom{N}{2} \Gamma_2^N \left(\left| \varphi_{l_1} \varphi_{l_2} \right\rangle \left\langle \varphi_{m_1} \varphi_{m_2} \right| \right) = \Gamma_2 \left(\left| \varphi_{l_1} \varphi_{l_2} \right\rangle \left\langle \varphi_{m_1} \varphi_{m_2} \right| \right) \qquad (57)
\end{aligned}
$$

In general one has

$$
\begin{aligned}
&a_{l_1}^{\dagger} \cdots a_{l_P}^{\dagger} a_{m_Q} \cdots a_{m_1} = \Gamma \left(\left| \varphi_{l_1} \cdots \varphi_{l_P} \right\rangle \left\langle \varphi_{m_1} \cdots \varphi_{m_Q} \right| \right) \\
&= \sum_{0 \leq N \leq \infty} \sum_{1 \leq j_1 < \cdots < j_N \leq \infty} \left| \varphi_{l_1} \cdots \varphi_{l_P} \varphi_{j_1} \cdots \varphi_{j_N} \right\rangle \left\langle \varphi_{m_1} \cdots \varphi_{m_Q} \varphi_{j_1} \cdots \varphi_{j_N} \right| \\
&= \sum_{0 \leq N \leq \infty} \sum_{1 \leq j_1, \cdots, j_N \leq \infty} \sqrt{(N+1) \cdots (N+P)} \sqrt{(N+1) \cdots (N+Q)} \left| \varphi_{l_1} \wedge \cdots \wedge \varphi_{l_P} \wedge \varphi_{j_1} \wedge \cdots \wedge \varphi_{j_N} \right\rangle \left\langle \varphi_{m_1} \wedge \right. \\
&= \sum_{0 \leq N \leq \infty} \sqrt{(N+1) \cdots (N+P)} \sqrt{(N+1) \cdots (N+Q)} \left| \varphi_{l_1} \right\rangle \left\langle \varphi_{m_1} \right| \wedge \cdots \wedge \left| \varphi_{l_P} \right\rangle \left\langle \varphi_{m_Q} \right| \wedge P_{\mathcal{H}^N} \\
&= \sum_{0 \leq N \leq \infty} \sqrt{\frac{(N+1) \cdots (N+P)}{P!}} \sqrt{\frac{(N+1) \cdots (N+Q)}{Q!}} \left| \varphi_{l_1} \cdots \varphi_{l_P} \right\rangle \left\langle \varphi_{m_1} \cdots \varphi_{m_Q} \right| \wedge P_{\mathcal{H}^N} \\
&= \sum_{0 \leq N \leq \infty} \sqrt{\frac{(N+1) \cdots (N+P)}{P!}} \sqrt{\frac{(N+1) \cdots (N+Q)}{Q!}} \Gamma_{PQ}^{N+P, N+Q} \left(\left| \varphi_{l_1} \cdots \varphi_{l_P} \right\rangle \left\langle \varphi_{m_1} \cdots \varphi_{m_Q} \right| \right) \\
&= \sum_{\min(P,Q) \leq N \leq \infty} \sqrt{\binom{N+P}{P} \binom{N+Q}{Q}} \Gamma_{PQ}^{N+P, N+Q} \left(\left| \varphi_{l_1} \cdots \varphi_{l_P} \right\rangle \left\langle \varphi_{m_1} \cdots \varphi_{m_Q} \right| \right) = \Gamma_{PQ} \left(\left| \varphi_{l_1} \cdots \varphi_{l_P} \right\rangle \left\langle \varphi_{m_1} \cdots \right. \right. \\
&\qquad (58)
\end{aligned}
$$

i.e.

$$
\begin{aligned}
&a^{\dagger}_{l_1}\cdots a^{\dagger}_{l_4}a_{m_2}a_{m_1} = \Gamma\left(\left|\varphi_{l_1}\cdots\varphi_{l_4}\right\rangle\left\langle\varphi_{m_1}\varphi_{m_2}\right|\right) = \sum_{0\leq N\leq\infty}\sum_{1\leq j_1<\cdots<j_N\leq\infty}\left|\varphi_{l_1}\cdots\varphi_{l_4}\varphi_{j_1}\cdots\varphi_{j_N}\right\rangle\left\langle\varphi_{m_1}\varphi_{m_2}\varphi_{j_1}\cdots\varphi_{j_N}\right. \\
&= \left|\varphi_{l_1}\cdots\varphi_{l_4}\right\rangle\left\langle\varphi_{m_1}\varphi_{m_2}\right| + \sum_{1\leq j_1<\cdots<j_N\leq\infty}\left|\varphi_{l_1}\cdots\varphi_{l_4}\varphi_{j_1}\right\rangle\left\langle\varphi_{m_1}\varphi_{m_2}\varphi_{j_1}\right| + \sum_{1\leq j_1<\cdots<j_N\leq\infty}\left|\varphi_{l_1}\cdots\varphi_{l_4}\varphi_{j_1}\varphi_{j_2}\right\rangle\left\langle\varphi_{m_1}\cdot\right. \\
&= \sum_{0\leq N\leq\infty}\sum_{1\leq j_1,\cdots,j_N\leq\infty}\sqrt{(N+1)\cdots(N+4)}\sqrt{(N+1)(N+2)}\left|\varphi_{l_1}\wedge\cdots\wedge\varphi_{l_4}\wedge\varphi_{j_1}\wedge\cdots\wedge\varphi_{j_N}\right\rangle\left\langle\varphi_{m_1}\wedge\varphi_{m_2}\right. \\
&= \sum_{0\leq N\leq\infty}\sqrt{(N+1)\cdots(N+4)}\sqrt{(N+1)(N+2)}\left|\varphi_{l_1}\wedge\cdots\wedge\varphi_{l_4}\right\rangle\left\langle\varphi_{m_1}\wedge\varphi_{m_2}\right|\wedge P_{\mathcal{H}^N} \\
&= \sum_{0\leq N\leq\infty}\sqrt{\frac{(N+1)\cdots(N+4)}{4!}}\sqrt{\frac{(N+1)(N+2)}{2!}}\left|\varphi_{l_1}\cdots\varphi_{l_4}\right\rangle\left\langle\varphi_{m_1}\varphi_{m_2}\right|\wedge P_{\mathcal{H}^N} \\
&= \sum_{0\leq N\leq\infty}\sqrt{\frac{(N+1)\cdots(N+4)}{4!}}\sqrt{\frac{(N+1)(N+2)}{2!}}\Gamma^{N+4,N+2}_{4,2}\left(\left|\varphi_{l_1}\cdots\varphi_{l_4}\right\rangle\left\langle\varphi_{m_1}\varphi_{m_2}\right|\right) \\
&= \left|\varphi_{l_1}\cdots\varphi_{l_4}\right\rangle\left\langle\varphi_{m_1}\varphi_{m_2}\right| + \sqrt{\frac{(5)\cdots(2)}{4!}}\sqrt{\frac{(3)(2)}{2!}}\Gamma^{5,3}_{4,2}\left(\left|\varphi_{l_1}\cdots\varphi_{l_4}\right\rangle\left\langle\varphi_{m_1}\varphi_{m_2}\right|\right) \\
&+ \sqrt{\frac{(6)\cdots(3)}{4!}}\sqrt{\frac{(4)(3)}{2!}}\Gamma^{6,4}_{4,2}\left(\left|\varphi_{l_1}\cdots\varphi_{l_4}\right\rangle\left\langle\varphi_{m_1}\varphi_{m_2}\right|\right) + \cdots \\
&= \left|\varphi_{l_1}\cdots\varphi_{l_4}\right\rangle\left\langle\varphi_{m_1}\varphi_{m_2}\right| + \sqrt{\binom{5}{4}}\sqrt{\binom{3}{2}}\Gamma^{5,3}_{4,2}\left(\left|\varphi_{l_1}\cdots\varphi_{l_4}\right\rangle\left\langle\varphi_{m_1}\varphi_{m_2}\right|\right) + \sqrt{\binom{6}{4}}\sqrt{\binom{4}{2}}\Gamma^{6,4}_{4,2}\left(\left|\varphi_{l_1}\cdots\varphi_{l_4}\right\rangle\left\langle\varphi_{m_1}\varphi\right.\right.
\end{aligned}
\tag{59}
$$

$$
\sum_{2\leq N\leq\infty}\sqrt{\binom{N+2}{4}}\sqrt{\binom{N}{2}}\Gamma^{N+2,N}_{4,2}\left(\left|\varphi_{l_1}\cdots\varphi_{l_4}\right\rangle\left\langle\varphi_{m_1}\varphi_{m_2}\right|\right) = \Gamma_{4,2}\left(\left|\varphi_{l_1}\cdots\varphi_{l_4}\right\rangle\left\langle\varphi_{m_1}\varphi_{m_2}\right|\right)
\tag{60}
$$

The components other than $\Gamma_{4,2}\left(\left|\varphi_{l_1}\cdots\varphi_{l_4}\right\rangle\left\langle\varphi_{m_1}\varphi_{m_2}\right|\right)$ are zero.

In general

$$
\begin{aligned}
\Gamma(X) = \Gamma\left(\sum_{0\leq P,Q\leq\infty}X_{PQ}\right) &= \sum_{0\leq P,Q\leq\infty}\Gamma(X_{PQ}) = \sum_{0\leq M,N\leq\infty}\Gamma_{MN}(X) \\
&= \sum_{0\leq M,N\leq\infty}\sum_{0\leq P\leq M,0\leq Q\leq N}\sqrt{\binom{M}{P}\binom{N}{Q}}\Gamma^{MN}_{PQ}(X_{PQ}).
\end{aligned}
\tag{61}
$$

i.e.

$$
\Gamma_{MN}(X) = \sum_{0\leq P\leq M,0\leq Q\leq N}\sqrt{\binom{M}{P}\binom{N}{Q}}\Gamma^{MN}_{PQ}(X_{PQ}).
\tag{62}
$$

One should note

$$\Gamma_{MN}\left(X_{PQ}\right)=\Gamma_{MN}\left(X_{PQ}\right)\delta_{M(N+P-Q)}=\Gamma_{N+PN+Q}\left(X_{PQ}\right)\delta_{M(N+P-Q)}=\sqrt{\binom{N+P}{P}\binom{N+Q}{Q}}\Gamma^{N+PN+Q}_{PQ}\left(X\right.$$

$$=\sum_{\substack{1\leq l_1<\cdots<l_p\leq\infty\\ 1\leq m_1<\cdots<m_Q\leq\infty}}\sqrt{\binom{N+P}{P}\binom{N+Q}{Q}}\Gamma^{N+PN+Q}_{PQ}\left(\left|\varphi_{l_1}\cdots\varphi_{l_P}\right\rangle\left\langle\varphi_{m_1}\cdots\varphi_{m_Q}\right|\right)X_{l_1\cdots l_P m_1\cdots m_Q}. \tag{63}$$

In Lowdin notation the basis transformation is

$$\Gamma\left|\left|\mathbf{\Phi}\right\rangle\left\langle\mathbf{\Phi}\right|\right)=\left|\left|\mathbf{\Phi}\right\rangle\left\langle\mathbf{\Phi}\right|\right)\mathcal{R}_{\varphi}\left(\Gamma\right)=\left|\mathbf{A}^{\dagger}\mathbf{A}\right)=\left|\mathbf{a}^{\dagger}\mathbf{a}\right)=\left|\Gamma\left(\left|\phi\right\rangle\left\langle\phi\right|\right),\Gamma\left(\left|\varphi_1\right\rangle\left\langle\phi\right|\right),\Gamma\left(\left|\phi\right\rangle\left\langle\varphi_1\right|\right),\cdots,\Gamma\left(\left|\varphi_P\right\rangle\left\langle\varphi_Q\right|\right),\right.$$

$$=\left|\Gamma_{00}\left(\left|\mathbf{\Phi}\right\rangle\left\langle\mathbf{\Phi}\right|\right),\cdots,\Gamma_{PQ}\left(\left|\mathbf{\Phi}\right\rangle\left\langle\mathbf{\Phi}\right|\right),\cdots\right), \tag{64}$$

where

$$\mathbf{a}^{\dagger}_{P}\mathbf{a}_{Q}=\sum_{0\leq M,N\leq r}\left|\varphi_M\right\rangle\left\langle\varphi_N\right|\Gamma_{\varphi MNPQ}=\sum_{0\leq M,N\leq r}\left|\varphi_M\right\rangle\left\langle\varphi_N\right|\mathcal{R}_{\varphi MNPQ}\left(\Gamma\right)=\Gamma_{PQ}\left(\left|\mathbf{\Phi}\right\rangle\left\langle\mathbf{\Phi}\right|\right). \tag{65}$$

Acting on the basis elements one obtains

$$\begin{aligned}
\mathbf{a}^{\dagger}_{P}\mathbf{a}_{Q}&=\Gamma\left(\left|\varphi_P\right\rangle\left\langle\varphi_Q\right|\right)=\Gamma_{PQ}\left(\left|\mathbf{\Phi}\right\rangle\left\langle\mathbf{\Phi}\right|\right)\\
&=\sum_{0\leq M,N\leq\infty}\sum_{\substack{1\leq j_1<\cdots<j_N\leq\infty\\ 1\leq k_1<\cdots<k_N\leq\infty}}\left|\left|\varphi_{j_1}\cdots\varphi_{j_M}\right\rangle\left\langle\varphi_{k_1}\cdots\varphi_{k_N}\right|\right)\Gamma_{j_1\cdots j_M k_1\cdots k_N l_1\cdots l_P m_1\cdots m_Q}\\
&=\sum_{0\leq N\leq\infty}\Gamma_{N+PN+Q}\left(\left|\left|\varphi_{l_1}\cdots\varphi_{l_P}\right\rangle\left\langle\varphi_{m_1}\cdots\varphi_{m_P}\right|\right)\right)\\
&=\sum_{0\leq N\leq\infty}\sqrt{\binom{N+P}{P}\binom{N+Q}{Q}}\Gamma^{N+P,N+Q}_{PQ}\left(\left|\varphi_{l_1}\cdots\varphi_{l_P}\right\rangle\left\langle\varphi_{m_1}\cdots\varphi_{m_Q}\right|\right)\\
&=\sum_{0\leq N\leq\infty}\sum_{1\leq j_1<\cdots<j_N\leq\infty}\left|\varphi_{l_1}\cdots\varphi_{l_P}\varphi_{j_1}\cdots\varphi_{j_N}\right\rangle\left\langle\varphi_{m_1}\cdots\varphi_{m_Q}\varphi_{j_1}\cdots\varphi_{j_N}\right|,
\end{aligned} \tag{66}$$

i.e.

$$\Gamma_{N+PN+Q}\left(\left|\left|\varphi_{l_1}\cdots\varphi_{l_P}\right\rangle\left\langle\varphi_{m_1}\cdots\varphi_{m_P}\right|\right)\right)=\sqrt{\binom{N+P}{P}\binom{N+Q}{Q}}\Gamma^{N+P,N+Q}_{PQ}\left(\left|\varphi_{l_1}\cdots\varphi_{l_P}\right\rangle\left\langle\varphi_{m_1}\cdots\varphi_{m_Q}\right|\right). \tag{67}$$

Γ

$$\mathcal{B}\left(\mathcal{H}^P,\mathcal{H}^Q\right)\to\mathcal{B}\left(\mathcal{H}^N,\mathcal{H}^M\right)$$

$$\Gamma^{MN}_{PQ}\left(\left|\varphi_{l_1}\cdots\varphi_{l_P}\right\rangle\left\langle\varphi_{m_1}\cdots\varphi_{m_Q}\right|\right)=\mathcal{A}_M X\otimes P_{\mathcal{H}^M,\mathcal{H}^N}\mathcal{A}_N=\left|\varphi_{l_1}\cdots\varphi_{l_P}\right\rangle\left\langle\varphi_{m_1}\cdots\varphi_{m_Q}\right|\wedge P_{\mathcal{H}^{N-P},\mathcal{H}^{M-Q}} \tag{68}$$

$\rfloor$

$\lceil$

Consider two bases $\{e_j\}$ and $\{f_j\}$ which are related by an invertible linear transformation M then

$$M\left|\mathbf{e}\right) = \left|\mathbf{e}\right)\mathcal{R}_e(M) = \left|\mathbf{f}\right) = \left|f_1, f_2, \cdots\right) = \left|M(e_1), M(e_2), \cdots\right) = \left|M_1(\mathbf{e}), M_2(\mathbf{e}), \cdots\right), \tag{69}$$

where

$$f_j = \sum_{1\le j\le r} e_k M_{kj} = M_j(\mathbf{e}) = M(e_j). \tag{70}$$

and

$$\begin{aligned} M\left|\mathbf{e}\right) &= \left|\mathbf{e}\right)\mathcal{R}_e(M) \\ \mathbb{M}_e &= \Delta_e \mathcal{R}_e(M) \\ \Delta_e^{-1}\mathbb{M}_e &= \mathcal{R}_e(M) \end{aligned} \tag{71}$$

$$\begin{aligned} M^{-1}\left|\mathbf{f}\right) &= \left|\mathbf{f}\right)\mathcal{R}_f\left(M^{-1}\right) \\ \mathbb{M}_f &= \Delta_f \mathcal{R}_f\left(M^{-1}\right) \\ \Delta_f^{-1}\mathbb{M}_f &= \mathcal{R}_f\left(M^{-1}\right) \end{aligned} \tag{72}$$

giving the following transformation for the orthonormalized bases

$$\begin{aligned} M\left|\mathbf{e}\right)\Delta_e^{-\frac{1}{2}} &= \left|\mathbf{e}\right)\Delta_e^{-\frac{1}{2}}\mathcal{R}_e(M) \\ \breve{\mathbb{M}}_e &= \mathcal{R}_e(M). \end{aligned}$$

$$\begin{aligned} M^{-1}\left|\mathbf{f}\right)\Delta_f^{-\frac{1}{2}} &= \left|\mathbf{f}\right)\Delta_f^{-\frac{1}{2}}\mathcal{R}_f\left(M^{-1}\right) \\ \breve{\mathbb{M}}_f^{-1} &= \mathcal{R}_f\left(M^{-1}\right). \end{aligned}$$

One can also write this as

$$\begin{aligned} M\left|\mathbf{e}\right) &= \left|\mathbf{f}\right)\Delta_e^{-1}\left(\mathbf{e}\right| \\ M^{-1}\left|\mathbf{f}\right) &= \left|\mathbf{e}\right)\Delta_f^{-1}\left(\mathbf{f}\right| \end{aligned} \tag{73}$$

and in the non orthonormal case as

$$\begin{aligned} M\left|\mathbf{e}\right)\Delta_e^{-\frac{1}{2}} &= \left|\mathbf{f}\right)\Delta_f^{-\frac{1}{2}}\Delta_e^{-\frac{1}{2}}\left(\mathbf{e}\right| \\ M^{-1}\left|\mathbf{f}\right)\Delta_f^{-1} &= \left|\mathbf{e}\right)\Delta_e^{-\frac{1}{2}}\Delta_f^{-\frac{1}{2}}\left(\mathbf{f}\right|. \end{aligned} \tag{74}$$

]

Linear Maps $M:\mathcal{B}\left(\mathcal{H}_{\mathcal{F}}\right)\to\mathcal{B}\left(\mathcal{H}_{\mathcal{F}}\right)$ in general are given by

$$M=\sum_{0\leq P,Q,M,N\leq\infty}\left|\left|\Phi_{P}\right\rangle\left\langle\Phi_{Q}\right|\right)\mathbb{M}_{PQMN}\left(\left|\Phi_{M}\right\rangle\left\langle\Phi_{N}\right|\right| \tag{75}$$

$$\sum_{0\leq P,Q,M,N\leq\infty}\left|\left|\Phi_{P}\right\rangle\left\langle\Phi_{Q}\right|\right)\mathbb{M}_{PQMN}\left(\left|\Phi_{M}\right\rangle\left\langle\Phi_{N}\right|\right|\sum_{0\leq R,S\leq\infty}\left|\left|\Phi_{R}\right\rangle\left\langle\Phi_{S}\right|\right)X_{RS} \tag{76}$$

$$M\left(X\right)=\sum_{0\leq P,Q\leq\infty}\sum_{0\leq M,N\leq\infty}\left|\left|\Phi_{P}\right\rangle\left\langle\Phi_{Q}\right|\right)\mathbb{M}_{PQMN}X_{MN}=\sum_{0\leq P,Q\leq\infty}\left|\left|\Phi_{P}\right\rangle\left\langle\Phi_{Q}\right|\right)\left(\mathbb{M}\mathbf{X}\right)_{PQ}=\sum_{0\leq P,Q\leq\infty}M_{PQ}\left(X\right), \tag{77}$$

where

$$X=\sum_{0\leq R,S\leq\infty}\left|\left|\varphi_{l_1}\cdots\varphi_{l_R}\right\rangle\left\langle\varphi_{m_1}\cdots\varphi_{m_S}\right|\right)X_{l_1\cdots l_R m_1\cdots m_S}. \tag{78}$$

The expansion map can be expressed as

$$\Gamma=\sum_{0\leq P,Q,M,N\leq\infty}\sum_{\substack{1\leq l_1<\cdots<l_p\leq\infty\\1\leq m_1<\cdots<m_Q\leq\infty\\1\leq j_1<\cdots<j_M\leq\infty\\1\leq k_1<\cdots<k_N\leq\infty}}\left|\left|\varphi_{l_1}\cdots\varphi_{l_P}\right\rangle\left\langle\varphi_{m_1}\cdots\varphi_{m_Q}\right|\right)\mathbf{\Gamma}_{l_1\cdots l_p m_1\cdots m_Q j_1\cdots j_M k_1\cdots k_N}\left(\left|\varphi_{j_1}\cdots\varphi_{j_M}\right\rangle\left\langle\varphi_{k_1}\cdots\varphi_{k_N}\right.\right. \tag{79}$$

and its action as

$$\begin{aligned}\Gamma\left(X\right)&=\sum_{0\leq P,Q\leq\infty}\sum_{0\leq M,N\leq\infty}\left|\left|\Phi_{M}\right\rangle\left\langle\Phi_{N}\right|\right)\mathbf{\Gamma}_{MNPQ}X_{PQ}=\sum_{0\leq M,N\leq\infty}\left|\left|\Phi_{M}\right\rangle\left\langle\Phi_{N}\right|\right)\left(\mathbf{\Gamma}\mathbf{X}\right)_{MN}=\sum_{0\leq M,N\leq\infty}\Gamma_{MN}\left(X\right)\\&\equiv\sum_{0\leq M,N\leq\infty}\sum_{\substack{1\leq l_1<\cdots<l_p\leq\infty\\1\leq m_1<\cdots<m_Q\leq\infty}}\Gamma_{l_1\cdots l_M m_1\cdots m_N}\left(X\right)\end{aligned} \tag{80}$$

and is defined on basis elements as

$$\begin{aligned}\Gamma\left(\left|\left|\Phi_{P}\right\rangle\left\langle\Phi_{Q}\right|\right)\right)&=\sum_{0\leq M,N\leq\infty}\left|\left|\Phi_{M}\right\rangle\left\langle\Phi_{N}\right|\right)\mathbf{\Gamma}_{MNPQ}=\sum_{0\leq M,N\leq\infty}\Gamma_{MN}\left(\left|\left|\Phi_{P}\right\rangle\left\langle\Phi_{Q}\right|\right)\right)=\sum_{0\leq N\leq\infty}\left|\left|\Phi_{P+N}\right\rangle\left\langle\Phi_{Q+N}\right|\right)\mathbf{\Gamma}\\&=\sum_{0\leq N\leq\infty}\sqrt{\binom{N+P}{P}\binom{N+Q}{Q}}\Gamma_{PQ}^{N+P,N+Q}\left(\left|\varphi_{l_1}\cdots\varphi_{l_P}\right\rangle\left\langle\varphi_{m_1}\cdots\varphi_{m_Q}\right|\right)\end{aligned} \tag{81}$$

$$=\sum_{0\leq N\leq\infty}\sqrt{\binom{N+P}{P}\binom{N+Q}{Q}}\sum_{1\leq j_1<\cdots<j_N\leq\infty}\left|\left|\varphi_{l_1}\cdots\varphi_{l_P}\varphi_{j_1}\cdots\varphi_{j_N}\right\rangle\left\langle\varphi_{m_1}\cdots\varphi_{m_q}\varphi_{j_1}\cdots\varphi_{j_N}\right.\right. \tag{82}$$

$$=\sum_{0\leq N\leq\infty}\Gamma_{N+PN+Q}\left(\left|\left|\Phi_{P}\right\rangle\left\langle\Phi_{Q}\right|\right)\right), \tag{83}$$

i.e.

$$\Gamma_{P+MQ+N}\left(\left|\left|\Phi_P\right\rangle\left\langle\Phi_Q\right|\right)\right) = \sqrt{\binom{N+P}{P}\binom{N+Q}{Q}}\Gamma_{PQ}^{N+P,N+Q}\left(\left|\Phi_P\right\rangle\left\langle\Phi_Q\right|\right)$$

$$= \sum_{1\leq j_1<\cdots<j_p\leq\infty} \left|\left|\varphi_{l_1}\cdots\varphi_{l_M}\varphi_{j_1}\cdots\varphi_{j_P}\right\rangle\left\langle\varphi_{m_1}\cdots\varphi_{m_N}\varphi_{j_1}\cdots\varphi_{j_P}\right|\right)\Gamma_{l_1\cdots l_M j_1\cdots j_P m_1\cdots m_N j_1\cdots}$$

$$= \sqrt{\binom{N+P}{P}\binom{N+Q}{Q}} \sum_{1\leq j_1<\cdots<j_p\leq\infty} \left|\varphi_{l_1}\cdots\varphi_{l_P}\varphi_{j_1}\cdots\varphi_{j_N}\right\rangle\left\langle\varphi_{m_1}\cdots\varphi_{m_Q}\varphi_{j_1}\cdots\varphi_{j}\right. \tag{84}$$

In particular

$$\Gamma\left(\left|\left|\varphi_{l_1}\cdots\varphi_{l_P}\right\rangle\left\langle\varphi_{m_1}\cdots\varphi_{m_P}\right|\right)\right) = \sum_{0\leq N\leq\infty}\sum_{\substack{1\leq j_1<\cdots<j_N\leq\infty\\1\leq k_1<\cdots<k_N\leq\infty}} \left|\left|\varphi_{j_1}\cdots\varphi_{l_N}\right\rangle\left\langle\varphi_{k_1}\cdots\varphi_{k_N}\right|\right)\Gamma_{j_1\cdots j_N k_1\cdots k_N l_1\cdots l_P m_1\cdots m_p}$$

$$= \sum_{P\leq N\leq\infty}\binom{N}{P}\sum_{1\leq j_1<\cdots<j_p\leq\infty}\left|\varphi_{l_1}\cdots\varphi_{l_P}\varphi_{j_1}\cdots\varphi_{j_{N-P}}\right\rangle\left\langle\varphi_{m_1}\cdots\varphi_{m_P}\varphi_{j_1}\cdots\varphi_{j_{N-P}}\right|$$

$$= \sum_{P\leq N\leq\infty}\Gamma_{NN}\left(\left|\left|\varphi_{l_1}\cdots\varphi_{l_P}\right\rangle\left\langle\varphi_{m_1}\cdots\varphi_{m_P}\right|\right)\right) = \sum_{P\leq N\leq\infty}\binom{N}{P}\Gamma_{PP}^{N,N}\left(\left|\varphi_{l_1}\cdots\varphi_{l_P}\right\rangle\left\langle\varphi_{m_1}\cdots\varphi_{m_P}\right|\right), \tag{85}$$

i.e. as $\left|\left|\varphi_{l_1}\cdots\varphi_{l_P}\right\rangle\left\langle\varphi_{m_1}\cdots\varphi_{m_P}\right|\right)$ is a basis element one has

$$\Gamma_{NN}\left(\left|\left|\varphi_{l_1}\cdots\varphi_{l_P}\right\rangle\left\langle\varphi_{m_1}\cdots\varphi_{m_P}\right|\right)\right) = \binom{N}{P}\Gamma_{PP}^{NN}\left(\left|\varphi_{l_1}\cdots\varphi_{l_P}\right\rangle\left\langle\varphi_{m_1}\cdots\varphi_{m_P}\right|\right), \tag{86}$$

while for a general X

$$\Gamma_{NN}\left(X\right) = \sum_{0\leq P\leq N}\binom{N}{P}\Gamma_{PP}^{NN}\left(X_P\right) \tag{87}$$

$$= \sum_{0\leq P\leq N}\binom{N}{P}\Gamma_{PP}^{NN}\left(\left|\varphi_{l_1}\cdots\varphi_{l_P}\right\rangle\left\langle\varphi_{m_1}\cdots\varphi_{m_P}\right|\right)X_{l_1\cdots l_P m_1\cdots m_p}. \tag{88}$$

The inverse of M is given by

$$\sum_{0\leq P,Q,R\leq\infty} M_{PQ}\left(M^{-1}\right)_{QR} = I_{\mathcal{B}(\mathcal{H}_\mathcal{F})} = \sum_{0\leq P,Q,R\leq\infty}\left(M^{-1}\right)_{PQ}M_{QR}. \tag{89}$$

In particular if

$$L\left(X_N\right) = \sum_{0\leq P\leq\infty} L_P\left(X_N\right) = \sum_{0\leq P\leq N\leq\infty}\binom{N}{P}L_N^P\left(X_N\right),$$

so if $X = X_N$ one has

$$X_N = L^{-1}L\left(X_N\right) = \sum_{0\leq P\leq\infty}\left(L^{-1}\right)_P^N L_N^P\left(X_N\right). \tag{90}$$

$$D_{PQj_1\cdots j_Pk_1\cdots k_Q} = \left(a_{j_1}^\dagger \cdots a_{j_P}^\dagger a_{k_P} \cdots a_{k_1} \middle| D \right)_{\mathcal{F}} = \left(\Gamma \left(|\varphi_{j_1} \cdots \varphi_{j_P}\rangle \left\langle \varphi_{k_1} \cdots \varphi_{k_Q} \right| \right) \middle| D \right)_{\mathcal{F}}$$
$$= \left(|\varphi_{j_1} \cdots \varphi_{j_P}\rangle \left\langle \varphi_{k_1} \cdots \varphi_{k_Q} \right| \middle| L(D) \right)_{\mathcal{F}} = L(D)_{j_1\cdots j_Pk_1\cdots k_Q}, \tag{91}$$

which defines the operator

$$D_{PQ} : \mathcal{H}^Q \to \mathcal{H}^P \tag{92}$$

by

$$D_{PQ} = \sum_{\substack{1\leq j_1<\cdots<j_P\leq\infty \\ 1\leq k_1<\cdots<k_Q\leq\infty}} L(D)_{j_1\cdots j_Pk_1\cdots k_Q} |\varphi_{j_1} \cdots \varphi_{j_P}\rangle \left\langle \varphi_{k_1} \cdots \varphi_{k_Q} \right|. \tag{93}$$

The contraction and expansion maps are linear and have components

$$P_{\mathcal{B}(\mathcal{H}^P)} L P_{\mathcal{B}(\mathcal{H}^N)} = L_{PN} = \binom{N}{P}^{-1} L_N^P; N \geq P$$
$$P_{\mathcal{B}(\mathcal{H}^N)} \Gamma P_{\mathcal{B}(\mathcal{H}^P)} = \Gamma_{NP} = \binom{N}{P}^{-1} \Gamma_P^N; N \geq P \tag{94a}$$

and thus are triangular maps.

One *should note* that in general

$$\Gamma(A)\Gamma(B) \neq \Gamma(A \wedge B) \neq \Gamma(AB), \tag{95}$$

so Γ is not an algebraic map.

Thus

$$\Gamma\left(\sum_{1\leq l\leq r} |\varphi_l\rangle \langle\varphi_l| \right) = \Gamma(P_{\mathcal{H}^1}) = P_{\mathcal{H}^1} \wedge \left\{ \sum_{0\leq N\leq r} (N+1) P_{\mathcal{H}^N} \right\} = \sum_{0\leq N\leq r} (N+1) P_{\mathcal{H}^1} \wedge P_{\mathcal{H}^N}$$
$$= \sum_{0\leq N\leq r} (N+1) P_{\mathcal{H}^{N+1}} = \sum_{1\leq N\leq r} N P_{\mathcal{H}^N} = P_{\mathcal{H}^1} \wedge \mathfrak{S}_1 \tag{96}$$

and

$$\Gamma(P_{\mathcal{H}^2}) = P_{\mathcal{H}^2} \wedge \left\{ \sum_{0\leq N\leq\infty} \frac{(N+1)(N+2)}{2} P_{\mathcal{H}^N} \right\} = \sum_{0\leq N\leq\infty} \frac{(N+1)(N+2)}{2} P_{\mathcal{H}^2} \wedge P_{\mathcal{H}^N}$$
$$= \sum_{0\leq N\leq\infty} \binom{N+2}{2} P_{\mathcal{H}^{N+2}} = \sum_{2\leq N\leq\infty} \binom{N}{2} P_{\mathcal{H}^N} = P_{\mathcal{H}^2} \wedge \mathfrak{S}_2, \tag{97}$$

where

$$\mathfrak{S}_2 = \sum_{0\leq N\leq\infty} \frac{(N+1)(N+2)}{2} P_{\mathcal{H}^N} = \sum_{0\leq N\leq\infty} \binom{N+2}{2} P_{\mathcal{H}^N} = |\phi\rangle\langle\phi| + 3P_{\mathcal{H}^1} + 6P_{\mathcal{H}^1}\wedge P_{\mathcal{H}^1} + 10P_{\mathcal{H}^1}\wedge P_{\mathcal{H}^1}\wedge P_{\mathcal{H}^1} + \tag{98}$$

A. Inverse

Note that

$$Y_N = \Gamma_N(X_P) = \sum_{0\leq P\leq N} \binom{N}{P} \Gamma_P^N(X_P) \tag{99}$$

and

$$Y_N = L_N(X_P) = \sum_{N\leq P\leq r} \binom{P}{N} L_P^N(X_P). \tag{100}$$

The inverse, Γ^{-1}, of the second quantization map is given by

$$Y_N = \Gamma_N^{-1}(X_P) = \sum_{0\leq P\leq N} (-1)^{N-P} \binom{N}{P} \Gamma_P^N(X_P) \tag{101}$$

and of the contraction map

$$Y_N = L_N^{-1}(X_P) = \sum_{N\leq P\leq r} (-1)^{P-N} \binom{P}{N} L_P^N(X_P). \tag{102}$$